

PRANAV SANGANI

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 PranavSangani
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EDUCATION

Purdue University, *Bachelor of Science in Computer Science – Junior*
Certificate in Entrepreneurship and Innovation

May 2028

EXPERIENCE

Axcellion, *Software Engineer Intern*

May 2026 – Present

- Built an end-to-end RAG pipeline in Python on AWS (S3, EC2) that parses and embeds 80K+ pages of FDA drug documents into a Neo4j knowledge graph.
- Raised search accuracy by over 50% by engineering hybrid retrieval on AWS, then extended it to EU drug data.
- Led development of a federated search API in Python that answers one query across FDA and EU data sources in ~0.3 seconds, with zero hallucinated answers in audit.
- Benchmarked several parsers and embedding models, choosing a stack that cut costs ~30x at 99.5% accuracy.
- Reorganized and consolidated company GitHub repositories, unifying 3 production pipelines into one codebase.

Aavalor Consulting, *Software Engineer Intern*

June 2026 – Sept 2026

- Co-built SHERPA, an AI assistant on the Claude Agent SDK and MCP that answers employee questions from company documents with citations, integrated with Microsoft Teams and SharePoint.
- Designed permission-aware document search in PostgreSQL so users only see files they are authorized to access, backed by 220+ automated tests running in CI on GitHub.
- Built a multi-agent evaluation pipeline in Python to measure answer accuracy, refusals, and access control.

DEEPSEAS (Drone Enabled Environment Patrol & Surveillance Edge AI System)

Oct 2020 – June 2024

- Co-developed a patent-pending underwater buoy that detects illegal fishing in Marine Protected Areas.
- Ran AI audio and vision models on a Jetson Nano edge device, trained on a curated dataset to identify vessel types.

GCL AI Summit - Capital One Hall

Sept 2022 – Aug 2023

- Co-led business operations for a charity AI summit for Title I students, featuring speakers from Meta and NVIDIA.
- Helped secure sponsorships from Google, Meta, Intel AI, and AWS; NBC-4 promotion drew around 300 students.

PROJECTS

[Sorting Algorithm Visualizer](#)

Nov 2024

- Built an interactive React app (JavaScript ES6+, HTML, CSS) that animates sorting algorithms in real time.
- Implemented merge, quick, insertion, and bubble sort with controls to switch algorithms and randomize arrays.

All-Pro NFL Predictor

July 2023

- Predicted NFL All-Pro selections in R using lasso regression, random forest, and neural network models.
- Wrote a Python web scraper that collected player statistics from Pro Football Reference as training data.

SKILLS

Languages

Python, C, SQL

Frameworks & Tools

AWS, Claude Agent SDK, MCP, PostgreSQL,
React, Git, GitHub

Certifications

Claude Certified Architect – Foundations
Anthropic | July 2026